

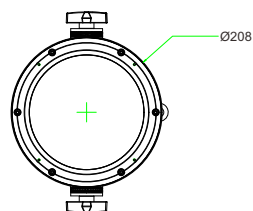
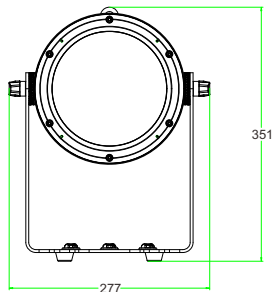
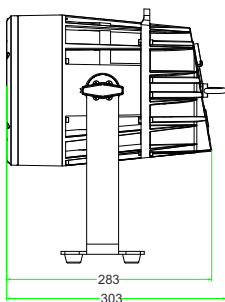
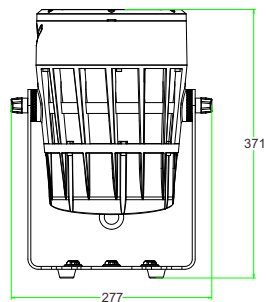
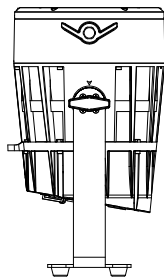
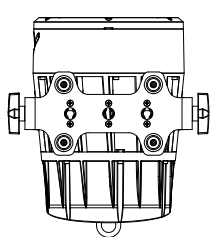


TABLE OF CONTENTS

Dimensions	1
Safety Instruction	2
Protection from burns and fire	3
Protection from injury	3
Fixture overview	4
Introduction	5
Powerful outdoor zoom par	5
Using for the first time	5
AC Power	5
Power voltage	5
Power cables	6
Data link	6
Tips for reliable data transmission	6
Physical installation	7
Fastening the fixture to a flat surface	7
Setup	8
Control panel and menu navigation	8
DMX address setting	8
Vision Control	8
Control mode	9
Error messages	9
Control panel	10
Dimmer mode	11
Onboard control menu	12/13
DMX protocol	14
2CH	14
4CH (CTC)	14
4CH (HSV)	14
5CH (basic)	15
5CH RGBW	15
5CH (Direct)	15
11CH (effect)	16
13CH (Advanced)	17
18CH (16-bit)	20/21
Specifications	22
Exploded view - 1	24
Exploded view - 2	25

DIMENSIONS

ALL DIMENSIONS ARE IN MILLIMETERS



SAFETY INSTRUCTION



WARNING!

Read the safety precautions in this section before installing, powering, operating or servicing this product.

The following symbols are used to identify important safety information on the product and in this manual:



DANGER!
Safety hazard.
Risk of severe injury or death.



DANGER!
Hazardous voltage. Risk of lethal or severe electric shock.



WARNING!
Fire hazard.



WARNING!
LED light emission. Risk of eye injury.



WARNING!
Burn hazard. Hot surface. Do not touch.



WARNING!
Wear protective eyewear.



WARNING!
Refer to user manual.

Risk group 2 (moderate risk) LED product according to EN 62471.

Warning! Do not look into the beam at a distance of less than 2 meters from the front surface of the product. Do not view the light output with optical instruments or any device that may concentrate the beam.



This product is for professional use only. It is not for household use.

This product presents risks of severe injury or death due to fire and burn hazards, electric shock and falls.

Read this manual before installing, powering or servicing the fixture, follow the safety precautions listed below and observe all warnings in this manual and printed on the fixture. If you have questions about how to operate the fixture safely, please contact your supplier.



PROTECTION FROM ELECTRIC SHOCK



- Disconnect the fixture from AC power before removing or installing any cover or part and when not in use.
- Always ground (earth) the fixture electrically.
- Use only a source of AC power that complies with local building and electrical codes and has both overload and ground-fault (earth-fault) protection.
- Before using the fixture, check that all power distribution equipment and cables are in perfect condition and rated for the current requirements of all connected devices.
- Power input and throughput cables must be rated 16A minimum, have three conductors 1.5 mm² (16 AWG) minimum conductor size and an outer cable diameter of 5 - 15 mm. Cables must be hard usage type (SJT or equivalent) and heat-resistant to 90° C minimum.
- Use only PowerCON TRUE 1® cable connectors to connect to power input sockets. Use only PowerCON TRUE 1® cable connectors to connect to power through put sockets.
- Isolate the fixture from power immediately if the power plug or any seal, cover, cable, or other component is damaged, defective, deformed, wet or showing signs of overheating. Do not reapply power until repairs have been completed.
- Refer any service operation not described in this manual to a qualified technician.
- Socket outlets used to supply fixture fixtures with power or external power switches must be located near the fixtures and easily accessible so that the fixtures can easily be disconnected from power.

PROTECTION FROM BURNS AND FIRE



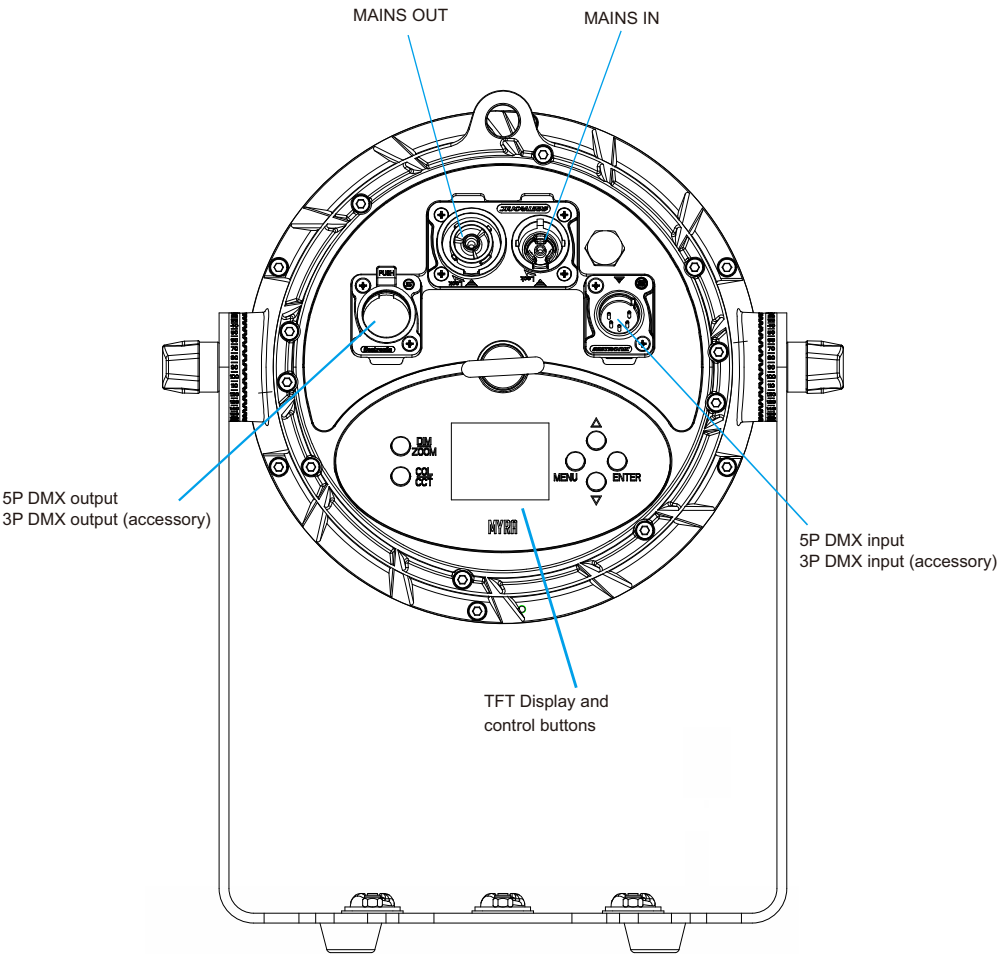
- The exterior of the fixture becomes hot during use. Avoid contact by persons and materials. Allow the fixture to cool for at least 5 minutes before handling.
- Keep all combustible materials (e.g. fabric, wood, paper) at least 300 mm away from the fixture.
- Keep flammable materials well away from the fixture.
- Ensure that there is free and unobstructed airflow around the fixture.
- Do not illuminate surfaces within 200 mm of the fixture.
- Do not attempt to bypass thermostatic switches or fuses.
- If you relay power from one fixture to another using power throughput sockets, do not connect more than ten fixture fixtures in total to each other in an interconnected chain.
- Connect only other fixture fixtures to fixture power throughput sockets.
- Do not connect any other type of device to these sockets.
- Do not stick filters, masks or other materials onto any optical component.
- Do not modify the fixture in any way not described in this manual.

PROTECTION FROM INJURY



- Do not look continuously at LEDs from a distance of less than 5 meters from the front surface of the fixture without protective eyewear such as shade 4-5 welding goggles. At less than this distance, the LED emission can cause eye injury or irritation. At distances of 5 meters and above, light output is harmless to the naked eye provided that the eye's natural aversion response is not overcome.
- Do not look at LEDs with magnifiers, telescopes, binoculars or similar optical instruments that may concentrate the light output.
- Ensure that persons are not looking at the LEDs from within 2 meters when the product lights up suddenly. This can happen when power is applied, when the product receives a DMX signal, or when SERVICE menu items are selected.
- Fasten the fixture securely to a fixed surface or structure when in use. The fixture is not portable when installed.
- Ensure that any supporting structure and/or hardware used can hold at least 10 times the weight of all the devices they support.
- Allow enough clearance around the head to ensure that it cannot collide with an object or another fixture when it moves.
- Check that all external covers and rigging hardware are securely fastened.
- Block access below the work area and work from a stable platform whenever installing, servicing or moving the fixture.
- Do not operate the fixture with missing or damaged covers, shields or any optical component.

FIXTURE OVERVIEW



INTRODUCTION

POWERFULL OUTDOOR ZOOM PAR

- Smooth RGB + Lime color mixing
- 15–45° electronic zoom range
- IP65 rating
- DMX512, RDM & Vision App control
- 1800–10000K variable color temperature

USING FOR THE FIRST TIME

Warning! Read “Safety Information” before installing, powering, operating or servicing the fixture. Before applying power to the fixture:

Thank you for purchasing our product. Before carrying out any operations with the product, carefully read this instruction manual and keep it with cure for future reference. It contains important information about the installation, usage and maintenance of the unit. Please strictly follow it.

The product conforms to the international standard DMX512 agreement. The products can be controlled in signal use and also can be controlled in many units. Sleek and stylish, precise and accurate optical system design, bring and clear light effects. All above makes our products applicable to be used within the scope of film and television, theater, live events, trade show exhibits, educational facilities, retail environments, and corporate shows, etc.

* The product is compliance with the CE & ROHS Standard.

* The product belongs to our company. The intellectual property is exclusively reserved. We will file a lawsuit on condition that it is illegally copied and imitated.

AC POWER



Warning! Read “Safety Information” starting on before connecting the fixtures to AC mains power.

Warning! For protection from electric shock, the fixture must be grounded (earthed). The power distribution circuit must be equipped with a fuse or circuit breaker and ground-fault (earth-fault) protection.

Warning! Socket outlets or external power switches used to supply the fixture with power must be located near the fixture and easily accessible so that the fixtures can easily be disconnected from power.



Important! Do not insert or remove live PowerCON TRUE 1 ® connectors to apply or cut power, as this may cause arcing at the terminals that will damage the connectors.

Important! Do not use an external dimming system to supply power to the fixture, as this may cause damage to the fixture that is not covered by the product warranty.

The fixture can be hard-wired to a electrical installation if you want to install it permanently, or a power plug that is suitable for the local power outlets can be installed on the power cable.

POWER VOLTAGE



Warning! Check that the voltage range specified on the fixtures serial number label matches the local AC mains power voltage before applying power to the fixture.

The fixtures accepts AC mains power at 100-240 V nominal, 50/60 Hz. Do not apply AC mains power to the fixture at any other voltage than specified.

POWER CABLES

Power input and throughput cables must be rated 16A minimum, have three conductors 1.5 mm² (16 AWG) minimum conductor size and an outer cable diameter of 5 - 15 mm. Cables must be hard usage type (SJT or equivalent) and heat- resistant to 90°C minimum. In the EU the cable must be HAR approved or equivalent.

If you install a power plug on the power cable, install a grounding-type (earthed) plug that is rated 16A minimum. Follow the plug manufacturer's instructions. Table 1 shows standard wire color-coding schemes and some possible pin identification schemes; if pins are not clearly identified.

Wire Color (EU models)	Wire Color (US models)	Conductor	Symbol
Brown	Black	Live	L
Blue	White	Neutral	N
Yellow/Green	Green	Ground (earth)	 or 

Table 1 : Wire color-coding and power connections

DATA LINK

A DMX 512 data link is required in order to control a fixture via DMX. The fixture has 5-pin XLR connectors for DMX data input and output. The pin-out on all connectors is pin 1 = shield, pin 2 = cold (-), and pin 3 = hot (+) Pins 4 and 5 in the 5-pin XLR connectors are not used.

TIPS FOR RELIABLE DATA TRANSMISSION

To connect the fixture to data:

1. Connect the DMX data output from the controller to the 5-pin XLR connector of the nearest fixture.
2. Connect the DMX output of the fixture closest to the controller to the DMX input of the next fixture and continue connecting fixtures output to input.

PHYSICAL INSTALLATION



Warning! The fixture must be either fastened to a flat surface such as a stage or wall, or clamped to a truss or similar structure in any orientation using a rigging clamp.

Warning! If the fixture can cause injury or damage if it falls, attach an approved safety cable to one of the safety cable attachment points on the base (see "Fixture overview").

Check that all surfaces to be illuminated are minimum 200 mm. from the fixture, that combustible materials (wood, fabric, paper, etc.) are minimum 100 mm. from the fixture, that there is free airflow around the fixture and that there are no flammable materials nearby.



FASTENING THE FIXTURE TO A FLAT SURFACE

The fixture can be fastened to a fixed flat surface that is oriented at any angle. Check that the surface can support at least 10 times the weight of all fixtures and equipment to be installed on it.

Warning! The supporting surface must be hard and flat or cooling may be blocked, which will cause overheating. Fasten the fixture securely. Do not stand it on a surface or leave it where it can be moved or can fall over. Attach a securely anchored safety cable to the safety cable attachment point (see "Fixture overview") if the fixture is to be installed in any location where it may fall and cause injury or damage if the primary attachment fails.

1. Block access under the work area. Working from a stable platform, hang the fixture on the truss with the arrow on the base towards the area to be illuminated. Tighten the rigging clamp.
2. Secure the fixture against clamp failure with a secondary attachment such as an approved safety cable that is rated for the weight of the fixture using one of the attachment points at the edges of the base (see "Fixture overview"). Do not use any other part of the fixture as a safety cable attachment point.

The installation of the fixture has to be built and constructed in a way that it can hold 10 times the weight for 1 hour without any harming deformation.

The installation must always be secured with a secondary safety attachment, e.g. an appropriate catch net. This secondary safety attachment must be constructed in a way that no part of the installation can fall down if the main attachment fails.

When rigging, derigging or servicing the fixture staying in the area below the installation place, on bridges, under high working places and other endangered areas is forbidden.

The operator has to make sure that safety-relating and machine-technical installations are approved by an expert before taking into operation for the first time and after changes before taking into operation another time.

The operator has to make sure that safety-relating and machine-technical installations are approved by an expert after every four year in the course of an acceptance test.

The operator has to make sure that safety-relating and machine-technical installations are approved by a skilled person once a year.

The fixture should be installed outside areas where persons may walk by or be seated.

IMPORTANT! OVERHEAD RIGGING REQUIRES EXTENSIVE EXPERIENCE, including (but not limited to) calculating working load limits, installation material being used, and periodic safety inspection of all installation material and the projector. If you lack these qualifications, do not attempt the installation yourself, but instead use a professional structural rigger. Improper installation can result in bodily injury or damage to property. The fixture has to be installed out of the reach of people.

SETUP

Warning! Read "Safety Information" before installing, powering, operating the fixture.

CONTROL PANEL AND MENU NAVIGATION

The onboard control panel and backlit graphic display are used to set the fixture's DMX address, configure individual fixture settings (personality), read out data and execute service utilities. See "Onboard control menus" for a complete list of menus and commands.

Using the control buttons

- To enter the menu select [MENU].
- Press [UP] and [DOWN] to scroll within a menu or adjust values.
- To enter a menu, select a function or apply a selection, press [ENTER].
- To escape a function or move back one level in the menu structure, press [MENU]
- To change fast dimmer and zoom, press [DIM/ZOOM].
- To change fast color and CCT levels, press [COL/CCT].

DMX ADDRESS SETTING

The DMX address, also known as the start channel, is the first channel used to receive instructions from the controller. For independent control, each fixture must be assigned its to a separate channel. The DMX address can be configured by using the DMX ADDRESS menu in the control panel. For setting the DMX address press [ENTER] before you can change the address.

- The main screen will show a  and the backlight will be switched off when a DMX signal is detected.
- The fixture is fully RDM ready. So when you are using a RDM ready console you can address the unit and read out its complete status. For RDM functions please refer to the ANSI/ESTA E1.20-2006 standard

VISION CONTROL (ONLY SHOWS UP WHEN VISION BOARD IS INSTALLED)

Go to the main menu, press the UP or DOWN buttons to find Vision APP and then click the ENTER button.

Then the main interface shows the APP in a red "unlinked" state, and the fixture also emit red light simultaneously. This indicates that the fixtures aren't in a networked state.

Open the "Vision" APP, click on the "create network" function. After the network is successfully created, add Tyra to the network. The main page will display the APP in a green "linked" state.

When multiple Tyra fixtures are connected to form a network, the Tyra with the strongest signal will display a blue "Master" on the screen, indicating that it is the master. The "master" will then emit blue light. The rest of the Tyra will display a green "slave" status, which indicates a slave unit.

When Tyra needs to exit the network that has been created, it can find "Unlink" on the Vision App page in the menu, and then click the ENTER button. The Tyra page will display a red "unlinked", which indicates that the fixture isn't in the network state.

CHECK REALITY

CONTROL MODE

DMX control mode is selected in the CONTROL MODE menu. The fixture can be controlled with 9 DMX control modes:

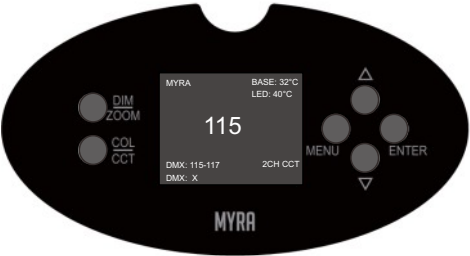
	2CH	4CH	4CH	5CH	5CH	5CH	11CH	13CH	18CH
		CTC	HSV	BASIC	RGBW	DIRECT	EFFECT	ADVAN.	16BIT
BLINDER	✓			✓			✓	✓	✓
CCT CONTROL		✓						✓	
16 BIT CONTROL				✓					✓
ZOOM CONTROL	✓	✓	✓				✓	✓	✓
FUNCTION CONTROL				✓				✓	✓
HSV			✓						
RGBL						✓	✓	✓	✓
RGBW					✓				

Error messages

Error	Decision condition	Detection	Severity
Base too cold	< -20 °C	Real-time monitoring	Normal
Base too hot	> 95 °C	Real-time monitoring	Normal
Fixture eeprom error	Read eeprom error	Power-on detection	Important
Fan error	< 1000 RPM	Power-on detection	Important
3D sensor error	3D sensor not found	Power-on detection	Normal
Head too cold	< -20 °C	Real-time monitoring	Important
Head too hot	> 95 °C	Real-time monitoring	Important
Wireless module lost	Wireless module not found	Power-on detection	Important
LED board lost	LED board not found	Power-on detection	Important
Communication unstable		Real-time monitoring	

Slave: Slave works automatically. When one fixture is set to AUTO or STATIC COLOR, the other fixture (connected via the XLR out cable) will follow the first unit's effect.

CONTROL PANEL



Here you can set all functions for the fixture.

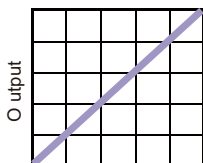
PERSONALITY	
FANS	Regulated = power : Min 65%, FAN : variable Full = power : Min 65%, FAN : 4900RPM Silent 1 = power : Min 50%, FAN : 3000RPM Silent 2 = power : Min 50% FAN : 2000RPM
Dimmer speed	"Normal" means select linear dimming, or choose dimmer 1-4 to control the dimming speed, dimming 1 of the fastest dimming curves, 3 for the most slowly dimming speed and halogen for a halogen effect dimming.
Dimmer curve	Linear / Square law / INV Square law/ S- Curve/Special
Bluetooth	Reset the Bluetooth board (if the board is installed)
LED Calibration	Factory calibration / User calibration / RAW
CRI	High / Low
Refresh rate	Controls the flicker frequency of the fixture 1200Hz / 2400Hz / 4000Hz / 6000Hz / 25000Hz
DMX HOLD	DMX HOLD = The fixture will remember on last values when you disconnect DMX NO DMX HOLD = The fixture has no output when you disconnect DMX
Display	Rotate: Auto= Auto Display Flip Function enabled Normal= Auto Display Flip Function disabled Background Light: always on = Display background light always on Auto off = Display deactivation after approximately 15second of inactivity BL blinking when no DMX : Yes = When there is no DMX, the DMX value will be blinking on main menu NO = When there is no DMX, the DMX value will be static on main menu
Indicator Light	Off = When the display is standby, no dot visible On= When the display is standby, there are 2 low brightness dots visible to show that the fixture is on and has DMX
KEY-Lock	Locks all the button functions. Standard unlocking password is (UP+DOWN+UP+DOWN+ENTER) Redshift
Alarm	On = create your own private password = Alarm is on (RDM alarm on is password 8888 to unlock) Off = enter your own private password = Alarm is off (Alarm is off (RDM alarm on is password 8888 to unlock)
Redshift	ON/OFF

Shortcut keys

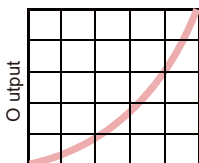
#	Shortcut
•	Press and hold MENU for the test function. Fixture will go back to normal working in 30seconds.
•	Press UP and DOWN at the same time for >3S seconds and the display will rotate.
•	Press and hold ENTER for >3S for the RDM QR Code (ELC).
•	Press UP and DOWN and ENTER at the same time for >3S to enter the secret menu.

DIMMER MODE

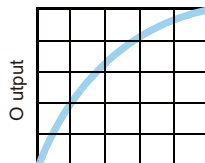
provides four dimming options (see picture below):



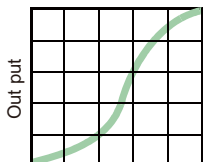
DMX %
LINEAR



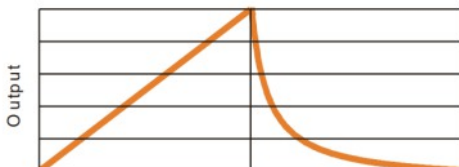
DMX %
SQUARE LAW



DMX %
INVERTED SQUARE LAW



DMX %
S - CURVE



DMX % UP
SPECIAL
DMX % Down

- **LINEAR** – the increase in light intensity appears to be linear as DMX value is increased.
 - **SQUARE LAW** – light intensity control is finer at low levels and coarser at high levels.
 - **INV Square law** – light intensity control is coarser at low levels and finer at high levels.
 - **S-CURVE** – light intensity control is finer at low levels and high levels and coarser at medium levels.
- Special** – the light intensity was linear increase with DMX value , and light intensity control is finer at low level with DMX values decrease , the dimmer speed will also has effect on it.

Whichever DIMMER CURVE option you select, you can choose between NORMAL or SMOOTH 1/2/ 3/ 4 dimming settings:

- **NORMAL** is the default setting. it gives a virtually instantaneous reaction when you dim from one intensity to another, but dimming slowly from one intensity to another may appear slightly uneven.
- **The SMOOTH 1/2/3/4/Special** setting gives smoother dimming during slow changes in intensity, but it limits the speed of dimming changes slightly. This makes it ideal for slow, smooth dimming, but a short time-lag may be noticeable if you try to dim quickly from one intensity to another.

ONBOARD CONTROL MENU

NO.	Main menu	Menu level 2	Menu level 3	Remark
1	DMX ADDRESS	- Set DMX Address - APP (only shows up when Bluetooth board is installed)	001 - 512 - APP (vition app control is on)	- Default: 001 - Default: Wire
2	CONTROL MODE	2CH 4CH (CTC) 4CH (HSV) 5CH (basic) 5CH RGBW 5CH (Direct) 11CH (effect) 13CH (Advanced) 18CH (16-bit)	1.Zoom, 2.Halogen/WW 1.Dimmer, 2.Zoom, 3.CCT, 4.CTC Tint 1.Zoom, 2.Hue, 3.Saturation, 4.Value 1.Red shift, 2.Dimmer curve, 3.Dimmer response, 4.Zoom, 5.Halogen/WW 1.Zoom, 2.Red, 3.Green, 4.Blue, 5.White (simulated) 1.Zoom, 2.Red, 3.Green, 4.Blue, 5.Lime 1.Dimmer, 2.Shutter, 3.Zoom, 4.Halogen/WW, 5.Red, 6.Green, 7.Blue, 8.Lime, 9.Preset color, 10.Auto, 11.Auto speed 1.Red shift, 2.Dimmer curve, 3.Dimmer response, 4.Control, 5.Dimmer, 6.Shutter, 7.Zoom, 8.Halogen/WW, 9.Red, 10.Green, 11.Blue, 12.CCT, 13.CTC tint 1.Red shift, 2.Dimmer curve, 3.Dimmer response, 4.Control, 5.Dimmer, 6.Dimmer fine, 7.Shutter, 8.Zoom, 9.Halogen/WW, 10.Halogen/WW fine, 11.Red, 12.Red fine, 13.Green, 14.Green fine, 15.Blue, 16.Blue fine, 17.Lime, 18.Lime fine	- Default: 11CH (effect)
3	PERSONALITY	- Fans - Dimmer Speed - Dimmer Curve - Bluetooth - Tungsten simulation - Calibration - CRI - Refresh rate - DMX hold - Display - Indicator light (only in DMX mode) - Key-Lock - Alarm - Font selection	- Regulated / Full / Silent 1 / Silent 2 - Normal / Smooth 1 / Smooth 2 / Halogen - Linear / Square law / INV Square law / S-Curve / Special - Reset the Bluetooth - On / Off - Factory calibration / User calibration / RAW - High / Low 1200 Hz / 2400 Hz / 4000 Hz / 6000 Hz / 25000 Hz - DMX hold / No DMX hold - Rotate: Auto / Manual - Background light: Always on / Auto off (15 s) - BL blinking when no DMX: Yes / No - Off / On (On = 2 dots showing Power + DMX always on) - Off / On Unlock: MENU + UP + MENU + DOWN + MENU + UP + MENU + DOWN + ENTER - Off / On Own password required - Helvetica / Arial / Univers / Calibri / Big Noodle Titling / Gill Sans / Book Antiqua	- Default: Regulated - Default: Normal - Default: Linear - Default: Off - Default: Factory - Default: Low - Default: 1200 Hz - Default: DMX hold - Default: Auto - Default: Auto off (15 s) - Default: Yes - Default: Off - Default: Off - Default: Off

NO.	Main menu	Menu level 2	Menu level 3	Remark
4	EDITOR	- Program 1 - Program 2 - Program 3	- Scene 1 ... Scene 30 (Pace time = Fade time + Wait time. When Pace time is 0, returns to first step.) - Dimmer: 0–255 Red: 0–255 Green: 0–255 Blue: 0–255 Lime: 0–255 Strobe: 0–255 - Fade time (M): 0–10 Fade time (S): 0–59 Fade time (mS): 0.00–0.99 - Wait time (M): 0–10 Wait time (S): 0–59 Wait time (mS): 0.00–0.99	
5	WIRELESS	- Receiver - Release	- On / Off (if Bluetooth Vision received is installed) - Release?	
6	STATIC COLOR	- Fixed color - CCT - Manual color - Strobe - Zoom - Dimmer	- R, G, B, L, RG, RB, GB, RGB, RGBL 2500 K, 2800 K, 3200 K, 4000 K, 5000 K, 5600 K, 6000 K, 7000 K, 8000 K, 9000 K, 10000 K - Red: 0–255 Green: 0–255 Blue: 0–255 Lime: 0–255 0–255 (0 = stop, 1–20 = slow @ fast) 0–255 0–255	- Default: RGBL - Default: 7000 K - Default: 0 - Default: 0 - Default: 0 - Default: 0
7	AUTO	- Auto - Speed - Dimmer	1 – 20 (Auto 1 ... Auto 20) 21 = Effect 1–20 loop play 0–255 (1–300 s per step) 0–255	- Default: 1 - Default: 0 - Default: 255
8	INFO	- Software type - Usage time - Temperature - FANS - RDM.UID - DMX LIVE - Error information - Wireless module (only when Bluetooth board is installed)	- Display board Vx.xx LED board Vx.xx IAP for D.B: Vx.xx Date: xx-xxxx - Total: xxxx hour LED total: xxxx h Power cycle: xxxxx times - Head xxx °C Stored max: xx °C Base xxx °C Stored max: xx °C - xxx RPM / xx % 0x02E2xxxxxxxx - Shows all received DMX data per chosen channel mode. All channels visible - Running normally / Error information... - Manufacturer: CRMX Module name: Timo RX Driver version: xxxxxxxx HW version: x.x.x.x Signal strength: xxx %	
9	FACTORY RESET	Factory reset?	Press ENTER Load.	

DMX PROTOCOL

2CH

2CH	Function	Value	Setting
1	Zoom	000-255	0 – 100%
2	Halogen / WW	000-255	0 – 100%

4CH (CTC)

4CH (CTC)	Function	Value	Setting
1	Dimmer	000-255	0 – 100%
2	Zoom	000-255	0 – 100%
3	CCT	000-009	No function
		010	1800 K
		025	2200 K
		040	2700 K — Bulb white
		055	3000 K
		070	3200 K — Halogen white
		085	4000 K — Neutral white
		100	4500 K
		115	5000 K
		130	5600 K — Studio white
		145	6000 K
		160	6500 K — Daylight white
		175	7000 K
		190	8000 K
		205-255	10000 K
4	CTC Tint	000	Neutral (0)
		001-127	–0.02 @ 0 (green shift)
		128	Neutral
		129-255	+0.02 (magenta shift)

4CH (HSV)

4CH (HSV)	Function	Value	Setting
1	Zoom	000-255	0 – 100%
2	Hue	000-255	0 – 100%
3	Saturation	000-255	0 – 100%
4	Value	000-255	0 – 100%

DMX PROTOCOL

5CH (basic)

5CH (basic)	Function	Value	Setting
1	Red shift	000-010	No function
		011-255	Red shift ON
2	Dimmer curve	000-090	No function
		091-095	Linear dimmer curve
		096-100	Square law dimmer curve
		101-105	INV square law dimmer curve
		106-110	S-Curve dimmer curve
		111-120	Special curve dimmer curve
3	Dimmer response	121-255	No use
		000-050	No function
		051-060	Dimmer speed normal
		061-070	Dimmer speed smooth 1
		071-080	Dimmer speed smooth 2
		081-090	Dimmer speed halogen
4	Zoom	091-255	No use
		000-255	0 – 100%
5	Halogen / WW	000-255	0 – 100%

5CH RGBW

5CH RGBW	Function	Value	Setting
1	Zoom	000-255	0 – 100%
2	Red	000-255	0 – 100%
3	Green	000-255	0 – 100%
4	Blue	000-255	0 – 100%
5	White (simulated)	000-255	0 – 100%

5CH (Direct)

5CH (Direct)	Function	Value	Setting
1	Zoom	000-255	0 – 100%
2	Red	000-255	0 – 100%
3	Green	000-255	0 – 100%
4	Blue	000-255	0 – 100%
5	Lime	000-255	0 – 100%

DMX PROTOCOL

11CH (effect) — default

Function		Value	Setting
11CH (effect)			
1	Dimmer	000-255	0 – 100%
		000-019	No function
		020-024	Shutter open
		025-064	Strobe 1 (fast – slow)
		065-069	Shutter open
		070-084	Strobe 2: opening pulse (fast – slow)
		085-089	Shutter open
		090-104	Strobe 3: closing pulse (fast – slow)
		105-109	Shutter open
		110-124	Strobe 4: random strobe (fast – slow)
		125-129	Shutter open
		130-144	Strobe 5: random opening pulse (fast – slow)
		145-149	Shutter open
		150-164	Strobe 6: random closing pulse (fast – slow)
		165-169	Shutter open
		170-184	Strobe 7: burst pulse (fast – slow)
		185-189	Shutter open
2	Shutter	190-204	Strobe 8: random burst pulse (fast – slow)
		205-209	Shutter open
		210-224	Strobe 9: sine wave (fast – slow)
		225-229	Shutter open
		230-244	Strobe 10: burst (fast – slow)
		245-255	Shutter open
3	Zoom	000-255	0 – 100%
4	Halogen / WW	000-255	0 – 100%
5	Red	000-255	0 – 100%
6	Green	000-255	0 – 100%
7	Blue	000-255	0 – 100%
8	Lime	000-255	0 – 100%
9	Preset color	000-010	No function
		011-255	Preset color (see Color macros table)

See next page

DMX PROTOCOL

11CH (effect)	Function	Value	Setting
10	Auto	000-020	No function
		021-030	Auto 1
		031-040	Auto 2
		041-050	Auto 3
		051-060	Auto 4
		061-070	Auto 5
		071-080	Auto 6
		081-090	Auto 7
		091-100	Auto 8
		101-110	Auto 9
		111-120	Auto 10
		121-130	Auto 11
		131-140	Auto 12
		141-150	Auto 13
		151-160	Auto 14
		161-170	Auto 15
		171-180	Auto 16
		181-190	Auto 17
		191-200	Auto 18
		201-210	Auto 19
		211-220	Auto 20
		221-255	No function
11	Auto speed	000-255	0 – 100%

13CH (Advanced)

13CH (Advanced)	Function	Value	Setting
1	Red shift	000-010	No function
		011-255	Red shift ON
2	Dimmer curve	000-090	No function
		091-095	Linear dimmer curve
		096-100	Square law dimmer curve
		101-105	INV square law dimmer curve
		106-110	S-Curve dimmer curve
		111-120	Special curve dimmer curve
		121-255	No use
3	Dimmer response	000-050	No function
		051-060	Dimmer speed normal
		061-070	Dimmer speed smooth 1
		071-080	Dimmer speed smooth 2
		081-090	Dimmer speed halogen
		091-255	No use

See next page

DMX PROTOCOL

13CH (Advanced)

13CH (Advanced)	Function	Value	Setting
4	Control (executes after 3 s)	000-010	No function
		011-020	Fan: Regulated
		021-030	Fan: Full
		031-040	Fan: Silent 1
		041-050	Fan: Silent 2
		051-060	Dimmer speed normal
		061-070	Dimmer speed smooth 1
		071-080	Dimmer speed smooth 2
		081-090	Dimmer speed halogen
		091-095	Linear dimmer curve
		096-100	Square law dimmer curve
		101-105	INV square law dimmer curve
		106-110	S-Curve dimmer curve
		111-120	Special curve dimmer curve
		121-130	Tungsten emulation ON
		131-140	Tungsten emulation OFF
		141-150	Factory calibration
		151-160	User calibration
		161-170	RAW calibration
		171-180	CRI High
		181-190	CRI Low
		191-200	Refresh rate: 1200 Hz
		201-210	Refresh rate: 2400 Hz
		211-220	Refresh rate: 4000 Hz
		221-229	Refresh rate: 6000 Hz
		230-239	Refresh rate: 25000 Hz
		240-249	All reset
5	Dimmer	250-255	No use
		000-255	0 – 100%

See next page

13CH (Advanced)	Function	Value	Setting
6	Shutter	000-019	No function
		020-024	Shutter open
		025-064	Strobe 1 (fast – slow)
		065-069	Shutter open
		070-084	Strobe 2: opening pulse (fast – slow)
		085-089	Shutter open
		090-104	Strobe 3: closing pulse (fast – slow)
		105-109	Shutter open
		110-124	Strobe 4: random strobe (fast – slow)
		125-129	Shutter open
		130-144	Strobe 5: random opening pulse (fast – slow)
		145-149	Shutter open
		150-164	Strobe 6: random closing pulse (fast – slow)
		165-169	Shutter open
		170-184	Strobe 7: burst pulse (fast – slow)
		185-189	Shutter open
		190-204	Strobe 8: random burst pulse (fast – slow)
		205-209	Shutter open
		210-224	Strobe 9: sine wave (fast – slow)
		225-229	Shutter open
		230-244	Strobe 10: burst (fast – slow)
		245-255	Shutter open
7	Zoom	000-255	0 – 100%
8	Halogen / WW	000-255	0 – 100%
9	Red	000-255	0 – 100%
10	Green	000-255	0 – 100%
11	Blue	000-255	0 – 100%
12	CCT	000-009	No function
		010	1800 K
		025	2200 K
		040	2700 K — Bulb white
		055	3000 K
		070	3200 K — Halogen white
		085	4000 K — Neutral white
		100	4500 K
		115	5000 K
		130	5600 K — Studio white
		145	6000 K
		160	6500 K — Daylight white
		175	7000 K
		190	8000 K
		205-255	10000 K
13	CTC Tint	000-000	Neutral (0)
		001-127	-0.02 @ 0 (green shift)
		128-128	neutral
		129-255	+0.02 (magenta shift)

DMX PROTOCOL

18CH (16-bit)

18CH (16-bit)	Function	Value	Setting
1	Red shift	000-010	No function
		011-255	Red shift ON
		000-090	No function
2	Dimmer curve	091-095	Linear dimmer curve
		096-100	Square law dimmer curve
		101-105	INV square law dimmer curve
		106-110	S-Curve dimmer curve
		111-120	Special curve dimmer curve
		121-255	No use
3	Dimmer response	000-050	No function
		051-060	Dimmer speed normal
		061-070	Dimmer speed smooth 1
		071-080	Dimmer speed smooth 2
		081-090	Dimmer speed halogen
		091-255	No use
4	Control (executes after 3 s)	000-010	No function
		011-020	Fan: Regulated
		021-030	Fan: Full
		031-040	Fan: Silent 1
		041-050	Fan: Silent 2
		051-060	Dimmer speed normal
		061-070	Dimmer speed smooth 1
		071-080	Dimmer speed smooth 2
		081-090	Dimmer speed halogen
		091-095	Linear dimmer curve
		096-100	Square law dimmer curve
		101-105	INV square law dimmer curve
		106-110	S-Curve dimmer curve
		111-120	Special curve dimmer curve
		121-130	Tungsten emulation ON
		131-140	Tungsten emulation OFF
		141-150	Factory calibration
		151-160	User calibration
		161-170	RAW calibration
		171-180	CRI High
		181-190	CRI Low
		191-200	Refresh rate: 1200 Hz
		201-210	Refresh rate: 2400 Hz
		211-220	Refresh rate: 4000 Hz
		221-229	Refresh rate: 6000 Hz
		230-239	Refresh rate: 25000 Hz
		240-249	All reset
		250-255	No use
5	Dimmer	000-255	0 – 100%
6	Dimmer fine	000-255	0 – 100%

See next page

DMX PROTOCOL

18CH (16-bit)	Function	Value	Setting
7	Shutter	000-019	No function
		020-024	Shutter open
		025-064	Strobe 1 (fast – slow)
		065-069	Shutter open
		070-084	Strobe 2: opening pulse (fast – slow)
		085-089	Shutter open
		090-104	Strobe 3: closing pulse (fast – slow)
		105-109	Shutter open
		110-124	Strobe 4: random strobe (fast – slow)
		125-129	Shutter open
		130-144	Strobe 5: random opening pulse (fast – slow)
		145-149	Shutter open
		150-164	Strobe 6: random closing pulse (fast – slow)
		165-169	Shutter open
		170-184	Strobe 7: burst pulse (fast – slow)
		185-189	Shutter open
		190-204	Strobe 8: random burst pulse (fast – slow)
		205-209	Shutter open
		210-224	Strobe 9: sine wave (fast – slow)
		225-229	Shutter open
		230-244	Strobe 10: burst (fast – slow)
		245-255	Shutter open
8	Zoom	000-255	0 – 100%
9	Halogen / WW	000-255	0 – 100%
10	Halogen / WW fine	000-255	0 – 100%
11	Red	000-255	0 – 100%
12	Red fine	000-255	0 – 100%
13	Green	000-255	0 – 100%
14	Green fine	000-255	0 – 100%
15	Blue	000-255	0 – 100%
16	Blue fine	000-255	0 – 100%
17	Lime	000-255	0 – 100%
18	Lime fine	000-255	0 – 100%

SPECIFICATIONS

Power	
Input voltage & rate	100-240V, 50/60Hz
Standby power	6,5W
MAX. Power(at nominal voltage 230V)	200W
Stable power(at nominal voltage 230V)	160W
Typical current (at nominal voltage 230V)	0,9A
Power Factor	0.96
Power plug type	Seetronic PowerCon True

Configuration	
LED color	RGBL
LED color temperature	1800K - 10000K
Quantity of LED	1*200W RGBL
Dimming frequency	1K - 25K Hz
Dimmer resolution	16bit

Optical	
Beam angle	16° - 45°

Photometric	
Output @5M	2750 lux

Heat management	
Cooling type:	Forced air (temperature-regulated, low noise)
MAX. Ambient temp (Ta max)	40°C
MIN. Ambient temp (Ta min)	-25°C

Menu	
Auto program	Yes
Static color	Yes
Manual calibration	Yes
Factory calibration	No
Strobe speed	0 - 20Hz
Random strobe	Yes

* PF = power factor. Measurements made at nominal voltage with all LEDs at full intensity. Allow for a deviation of +/- 10%.

Control

Control protocol	USITT DMX512/1990
DMX channel range	2CH / 4CH (CTC) / 4CH (HSV) / 5CH / 5CH / 5CH / 11CH / 13CH / 18CH
RDM	Yes
RDM compliance	ANSI/ESTA E.120
CRMX	No
Vision APP ACN	Optional
DMX input connection	DMX 5P in & out (3 pin possible with optional chassis part)
Data input (artnet, SACN)	None

Hardware

Interface	Backlit TFT display
Software upload method	XLR via special box

Installation

IP rating	
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Housing

	IP65
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Safety attachment point	Yes
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Physical

Net product weight	5,26 kg
Machine dimensions - Length	277mm
Machine dimensions - Width	217mm
Machine dimensions - Height	371mm

Accessories

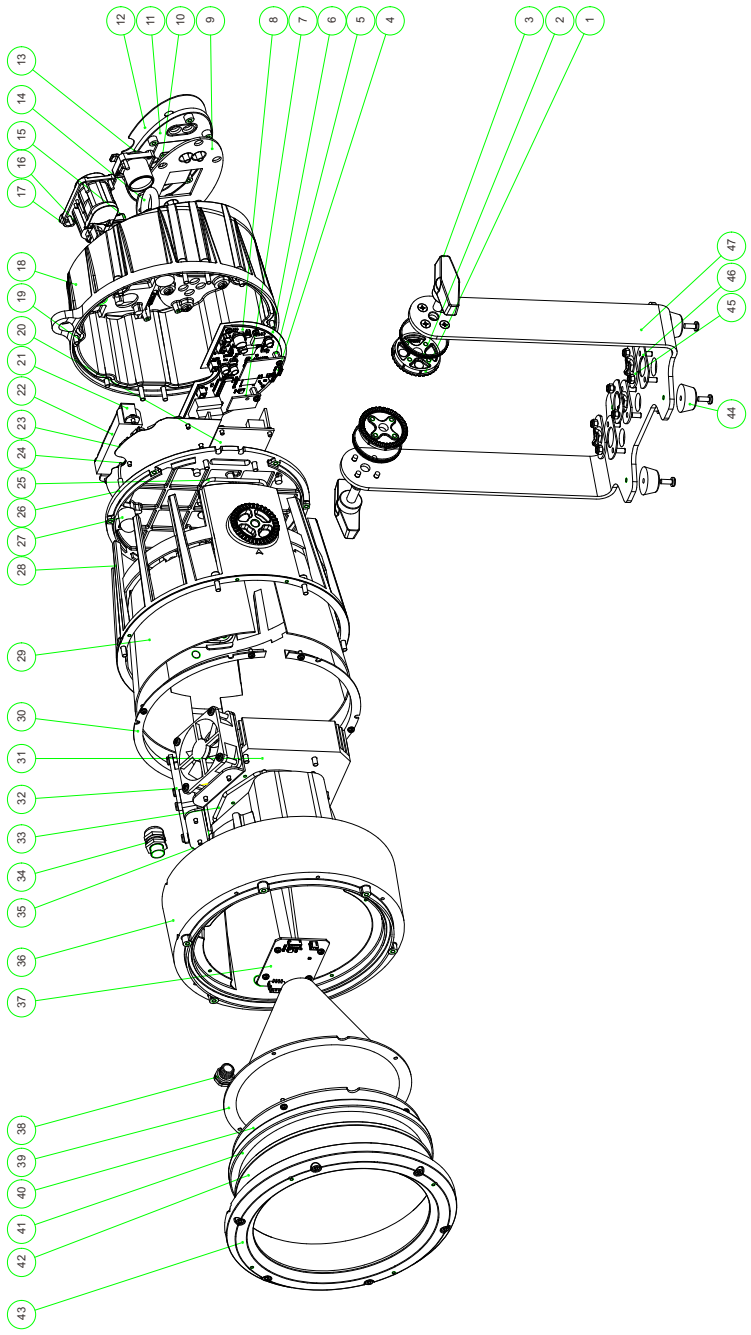
Included items	
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Approvals

	Power cable, manual
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Approved certifications	CE/ROHS
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EXPLODED VIEW



EXPLODED VIEW

NO.	Description	Part Number
1	Parts: CLF Nero Top cover	CLF-31-008
2	Parts: CLF Nero Tempered front glass	CLF-31-029
3	Parts: CLF Nero Front glass rubber pad	CLF-31-018
4	Parts: CLF Nero RGB LED board A	CLF-31-044A
5	Parts: CLF Nero Thermal insulation pad	CLF-31-035
6	Parts: CLF Nero LED board holder plate	CLF-31-017
7	Parts: CLF Nero LED board holder plate	CLF-31-016
8	Parts: CLF Nero Pressure plate insulation pad	CLF-31-038
9	Parts: CLF Nero Strobe lens	CLF-31-014
10	Parts: CLF Nero Strobe LED board	CLF-31-043
11	Parts: CLF Nero Strobe led board support radiator	CLF-31-015
12	Parts: CLF Nero Radiator	CLF-31-007
13	Parts: CLF Nero Bearing shaft rubber pad	CLF-31-023
14	Parts: CLF Nero Bearing seat rubber pad	CLF-31-022
15	Parts: CLF Nero Left axis support	CLF-31-026
16	Parts: CLF Nero large pulley support	CLF-31-028
17	Parts: CLF Nero heavy duty belt	CLF-31-048
18	Parts: CLF Nero Side arm cover rubber pad (left & right)	CLF-31-020
19	Parts: CLF Nero Arm outer cover (left & right)	CLF-31-011
20	Parts: CLF Nero Tilt motor + position sensor	CLF-31-037
21	Parts: CLF Nero Arm inner cover (left & right)	CLF-31-010
22	Parts: CLF Nero Side arm (arm -> base) waterproof rubber pad	CLF-31-021
23	Parts: CLF Nero Base rubber pad	CLF-31-019
24	Parts: CLF Nero Base bottom cover	CLF-31-013

NO.	Description	Part Number
25	Parts: CLF Nero Rubber feet	CLF-31-050
26	Parts: CLF Nero Base bottom plate + quick lock	CLF-31-031
27	Parts: CLF Nero Power supply	CLF-31-042
28	Parts: CLF Nero Fan fixed plate	CLF-31-034
29	Parts: CLF Nero Power supply fan	CLF-31-039
30	Parts: CLF Nero Right axis support	CLF-31-027
31	Parts: CLF Nero Bearing support	CLF-31-025
32	Parts: CLF Nero Quick lock adapter	CLF-31-051
33	Parts: CLF Nero rubber seal upgrade port	CLF-31-052
34	Parts: CLF Nero Waterproof RJ45 connector with waterproof cover	CLF-31-006
35	Parts: CLF Nero Display fixed support plate	CLF-31-032
36	Parts: CLF Nero Display board	CLF-31-046
37	Parts: CLF Nero Tempered display glass	CLF-31-030
38	Parts: CLF Nero Display rubber pad	CLF-31-024
39	Parts: CLF Nero Display board support plate	CLF-31-033
40	Parts: CLF Nero Waterproof 5 pin female dmx socket	CLF-31-005
41	Parts: CLF Nero Waterproof 5 pin male dmx socket	CLF-31-004
42	Parts: CLF Nero Powercon TRUE1 input socket with rubber pad	CLF-31-001
43	Parts: CLF Nero Rubber cover for DMX in	CLF-31-003
44	Parts: CLF Nero switching board insulation pad	CLF-31-036
45	Parts: CLF Nero Screw 1	Screw 1 Nero
46	Parts: CLF Nero screw 1 support	Screw support
47	Parts: CLF Nero Screw 2	Screw 2 Nero
48	Parts: CLF Nero RGB LED board B	CLF-31-045
49	Parts: CLF Nero Head fan	CLF-31-040
50	Parts: CLF Nero Base housing	CLF-31-012
x	Parts: CLF Nero ArtNet board	CLF-31-049

